NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	30/12/2025	Issued For Review	MLW	AS	RTN		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

TABEL REVISI

Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

No.	Page	Company's Comment	Contractor's Response	Remarks

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1. GENERAL

1.1 Latar Belakang

Struktur Tanjung Miring Barat (TMB) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 20 kilometer di sebelah tenggara Kota Prabumulih, dengan luas daerah sekitar 8 km². Struktur Karangan (KRG) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 15 kilometer di sebelah selatan Kota Prabumulih, dengan luas daerah sekitar 45 km².

ID	Easting (Meter)	Northing (Meter)	Elevation (Meter)	Feature Code
PN-1531	424962,745	967376,210	10,414	
PM-8593	425000,093	907528,777	19,046	

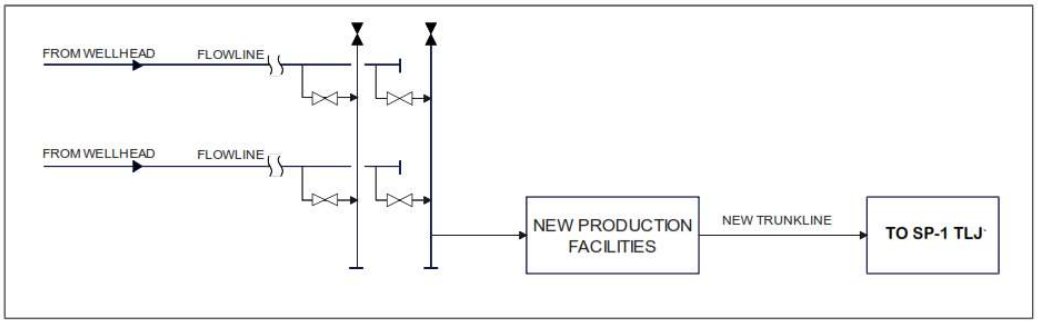
Gambar 1. Lokasi Rencana SP Tanjung Miring Barat - Karangan Timur

Lapangan Tanjung Miring Barat telah diproduksi sejak tahun 1938 dan terus beroperasi hingga tahun 2003 dengan produksi terakhir dari sumur TMB-11. Sejak dinonaktifkannya lapangan ini, fasilitas produksi yang terdiri dari flowline, stasiun pengumpul, serta jalur pengiriman pipa minyak ke custody transfer (PPP Prabumulih) sudah tidak layak operasi. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Baral serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah. Saat ini status SP TMB adalah idle atau tidak beroperasi, sehingga produksi sumuran menggunakan Tank On-Site (TOS) dan dikirim secara gross ke SP-II TMT menggunakan trucking. Dikarenakan tidak adanya line gas ke sumur-sumur struktur TMB, supply Listrik untuk pumping unit menggunakan solar (diesel engine). Lapangan Karangan Timur hanya diproduksikan dari 2 sumur yaitu KRG-9 dan KRG-10 beroperasi mulai tahun 1984, dimana kondisi eksisting jalur pengiriman minyak dari sumur menggunakan trucking. Apabila akan dilakukan

pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Barat serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah.

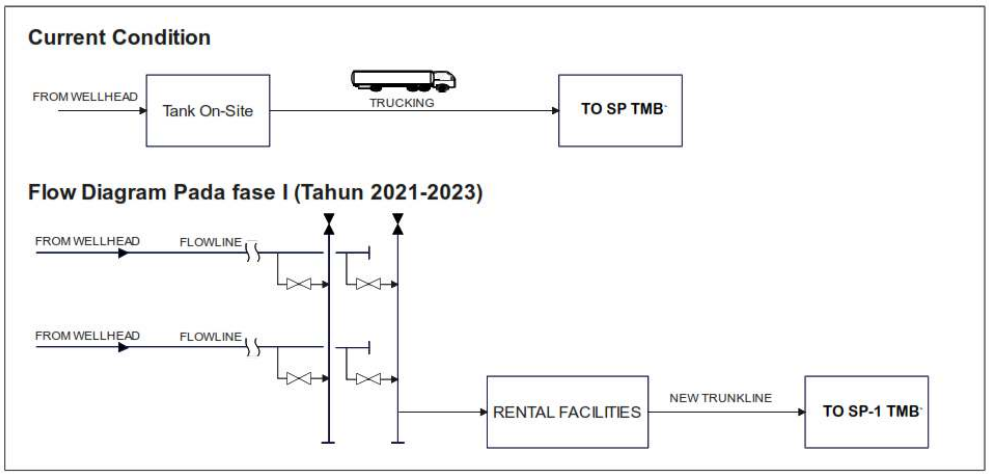
Dalam rencana jangka panjang sebagai upaya peningkatan produksi, PT Pertamina EP berencana akan mengembangkan Lapangan Tanjung Miring Barat dan Karangan Timur melalui pengembangan fasilitas produksi dimana rencana kerja pengembangan fasilitas akan dibagi ke dalam dua fase.

Pada fase pertama, produksi dari pengembangan awal TMB -KRG Timur akan dikirimkan menuju fasilitas sewa Early Production Facility (EPF) untuk selanjutnya dikirimkan menuju SP-1 TLJ melalui trunkline baru. Pada fase ini lingkup fasilitas produksi yang akan dibangun adalah flow/ine dari sumuran, sewa EPF dan trunkline dari EPF sampai dengan SP-1 TLJ.



Gambar 2. Rencana Pola Operasi Fase

Pada fase kedua, akan dilakukan pengembangan tambahan dari sumur-sumur TMB-KRG Timur. Pada fase ini, lingkup fasilitas produksi yang akan dibangun meliputi pembangunan flowline dari sumuran pengembangan lanjutan serta peningkatan fasilitas SP TMB untuk menggantikan sewa EPF.



Gambar 3. Rencana Pola Operasi Fase

1.2 Tujuan

Dokumen ini menjelaskan dipersiapkan sebagai Mechanical Datasheet for Storage Tank KRG (T-006 A/B) sebagai bagian dari desain yang harus diikuti oieh KONTRAKTOR dan VENDOR untuk proyek Pembangunan Fasilitas Stasiun Pengumpul Lapangan Tanjung Miring Barat- Karangan Timur, Field Prabumulih.

1.3 Ruang Lingkup

Mechanical Datasheet ini meliputi ketentuan minimal dalam mengatur desain. material,pabrikasi, pemeriksaan, pengujian, persiapan, pengecatan dan persiapan untuk pengiriman Storage Tank KRG (T-006 A/B) untuk dipasang di proyek Pembangunan Fasilitas Stasiun Pengumpul Lapangan Tanjung Miring Barat — Karangan Timur, Field Prabumulih.

1.4 Istilah

Dokumen ini berisi istilah-istilah yang memiliki definisi tercantum sebagai berikut:

Perusahaan	Perusahaan yang dimaksud adalah PT PERTAMINA HULU ROKAN selaku Pemilik dan Pengguna Fasilitas akhir.
Kontraktor	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan pengadaan dan konstruksi yang akan bertanggung jawab kepada PERUSAHAAN.
Vendor	Pihak lain yang akan menyediakan sesuatu untuk kebutuhan proyek, selain yang disediakan oleh PERUSAHAAN dan KONTRAKTOR, disebut VENDOR. Tanggung jawabnya langsung kepada KONTRAKTOR baik terhadap PO maupun bentuk perjanjian lain antara VENDOR dan

	KONTRAKTOR.
SKK MIGAS	Suatu badan dari Pemerintah Indonesia yang bertanggung jawab terhadap proses persetujuan dan izin untuk semua fasilitas di lingkungan minyak dan gas.
Perencana	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan Jasa FEED Engineering Services. Dalam hal ini disebut PERENCANA, yang akan bertanggung jawab kepada PERUSAHAAN sebagaimana dirinci dalam kontrak, scope of work, atau work order.
Inspektor	Perwakilan dari PERUSAHAAN dan/atau seseorang yang ditunjuk oleh PERUSAHAAN, yang bertanggung jawab sebagai INSPEKTOR atas pekerjaan yang dilakukan oleh KONTRAKTOR.SINGKATAN

2. KODE, STANDAR, DAN REFERENSI

DOKUMEN PROYEK

1. TMBP-SP001-ME-DBS-D101 - Mechanical Design Basis
2. TMBP-SP001-PR-PID-106 - P&ID Storage Tank KRG (T-006 AB)
3. TMBP-SP001-PR-DST-D115 - Process Datasheet - Storage Tank KRG (T-006 AB)
4. TMBP-SP001-ME-SPE-D102 - Specification for Tank (Tank API 650, FWKO, Wash Tank, etc.)

3. Daftar Singkatan

Beberapa singkatan yang paling umum digunakan tercantum dalam daftar berikut:

BOPD	*Barrel Oil Per Day*
BPD	*Barrel Per Day*
BWPD	*Barrel Water Per Day*
EOR	*Enhanced Oil Recovery*
FEED	*Front End Engineering Design*
KRG	*Karangan*
MMSCFD	*Million Standard Cubic Feet per Day*
NPS	*Nominal Pipe Size*

PB	*Production Battery*
PHR	*Pertamina Hulu Rokan*
PT	*Perseroan Terbatas*
SP	*Stasiun Pengumpul*
SCH	*Schedule*
TMB	*Tanjung Miring Barat*
TMT	*Tanjung Miring Timur*
WK	*Wilayah Kerja*

MECHANICAL DATASHEET FOR STORAGE TANK KRG (T-006 A)

Judul Proyek	Nomor Dokumen	Rev.	Date	Page
JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR	TMBP-SP001-ME-DST-D112	A	29-Dec-25	10 of 15

Tag No.	:	T-006 A	Client	:	PT. PERTAMINA EP
Service	:	STORAGE TANK KRG	Plant Location	:	SP TANJUNG MIRING
P & ID No.	:	TMBP-SP001-PR-PID-D106	No. of Unit	:	1 (ONE)
Eq. Layout No.	:	-			

REVISION

1	**DESIGN DATA**			**MATERIAL OF CONSTRUCTION**	
2	Fluid	GROSS FLUID (WATER+OIL)		Shell	A283 Gr. C (Note 25)
3	Code	API STD 650 Latest Ed		Roof & Bottom	A283 Gr. C (Note 25)
4	Capacity (Normal/Working)	2812.98 / 1100.00	bbls	Reinforcing Pads	A283 Gr. C
5	Operating Pressure	0,15	psig	Nozzle Neck - Pipe	A106 Gr.B
6	Operating Temp.	90	°F	Manhole Neck	A283 Gr. C
7	Design Pressure (max.)	Full of water + 1	psig	Manhole Flange	A283 Gr. C
8	Design Pressure (min.)	Vacuum - 0.06	psig	Manhole Cover	A283 Gr. C
9	Design Temperature (max./min.)	150/20	°F	Nozzle Flanges / Blinds	A105
10	Specific Gravity	0,950		Internal Pipe	A106 Gr.B
11	Flowrate IN/OUT	5055/1200	bpd	Coil Heater	A106 Gr.B
12	Type of Tank	Cylindrical Tank		Nozzle Fittings	A234 Gr.WPB
13	Tank Diameter (ID)	31.5' - 0" (9600)	ft-in (mm)	Bracket Supports	A283 Gr. C
14	Tank Height	25.3' - 0" (7700)	ft-in (mm)	Roof Structure	A36 / SS400
15	Corrosion Allowance			Earthing Lugs	SS 304
16	Shell/Roof/Bottom	0.125	in	Name Plates & Bracket	SS 316 & A283 Gr. C
17	Joint Efficiency	0,85		Ladder, Handrail, Platform, Circular Stair	A36 / SS400
18	Hydrotest Pressure	Full of Water		Anchor Bolt Chairs	A283 Gr. C
19	Wind Load:			Anchor (By Others) Bolts (Note 6)	A307 Gr. B Hot Dip Galvanized
20	Wind Design	ASCE 7-16		Nuts (Note 6)	A63 Gr. A Hot Dip Galvanized
21	Wind Speed	64,44	mph	Gaskets	**(Note 3)**
22	Exposure	C		Structural Bolts & Nuts	A325 (Hot Dip Galvanized)
23	Importance Factor	1,00		External Bolts / Nuts	A193 Gr. B7 / A194 Gr. 2H **(Note 4)**
24	Seismic Load:	API-650 - ASCE Mapped (Sa & S1)		Internal Bolts / Nuts	A193 Gr. B7 / A194 Gr. 2H **(Note 4)**
25	Seismic Factor	Sa=0.0422 g; S1=0.0627 g; PGA: 0.017 g		Stiffening Ring / Wind Girder	A283 Gr. C / A36 / SS400
26	Class	As per Soil Investigation		Vortex Breaker	A283 Gr. C
27	Importance Factor	1,5			
28	Heaters	N/A			
29	Heating Fluid	N/A			
30	Design Pressure	N/A	psig	**CONSTRUCTION**	
31	Design Temperature (Inlet/Outlet)	N/A	°F	Roof Type	Supported Cone Roof (1:6)
32	Coil Size & Thickness	N/A		Bottom Type	Slope Bottom (1:120)
33	Coil Surface Area	N/A	ft^2	Walkway / Stairway / Pipe Clip	Required
34	Coil minimum length	N/A	ft	Roof Drain System	Not Required
35	**INSPECTION, TESTING & CERTIFICATE**			Vent System	Required
36	Non Destructive Testing:			Firewater Ring and Clip Support	Required
37	Radiography	Yes, as per Code & Spec.		Foam Chamber	Required
38	Ultrasonic	Yes, as per Code & Spec.		Clips and Support	Required
39	Magnetic Partlicle	Yes, as per Code & Spec.		Earthing Boss	Required (2 ea)
40	Dye Penetrant	Yes, as per Code & Spec.		Internal Parts	
41	Hydrotest	As per Code & Spec.		- Vortex Breaker	Required
42	Post Weld Heat Treatment	As per Code & Spec. (if required)		- Internal Pipe	Required
43	Vacuum Test	As per Code & Spec.		- Foam Deflector	Required
44	Material Certification	Yes, Required Type 3.1		- Measurement Table	Required
45	Leak Test /Soap Bubble Test	Yes at 15 psig		- Draw Off Sump	Required
46	MIGAS Certified	Required		**LIQUID LEVEL**	
47	Calibration by DIMET	Required		HHLL (High High Liquid Level)	14.93' ft
48				HLL (High Liquid Level)	14.43' ft
49				LLL (Low Liquid Level)	7.0' ft
50	**PROTECTIVE COATING DATA**			LLLL (Low Low Liquid Level)	2.3' ft
51	Surface Praration Painting external	Yes (Note 5)			
52	internal	Yes (Note 25)		**ESTIMATED WEIGHTS (NOTE 23)**	
53	Insulation	N/A		Empty	59416 lb 26950 kg
54	Fireproofing	N/A		Operating	1158972 lb 525700 kg
55	Cathodic Protection	Yes (Note 15)		Hydrotest	1290280 lb 585260 kg
56	Type / Material / Weight	VTA / VTA / VTA			
57					

Any controlled copies may be superseded. Refer to COMPANY Server for controlled version.

REVISION

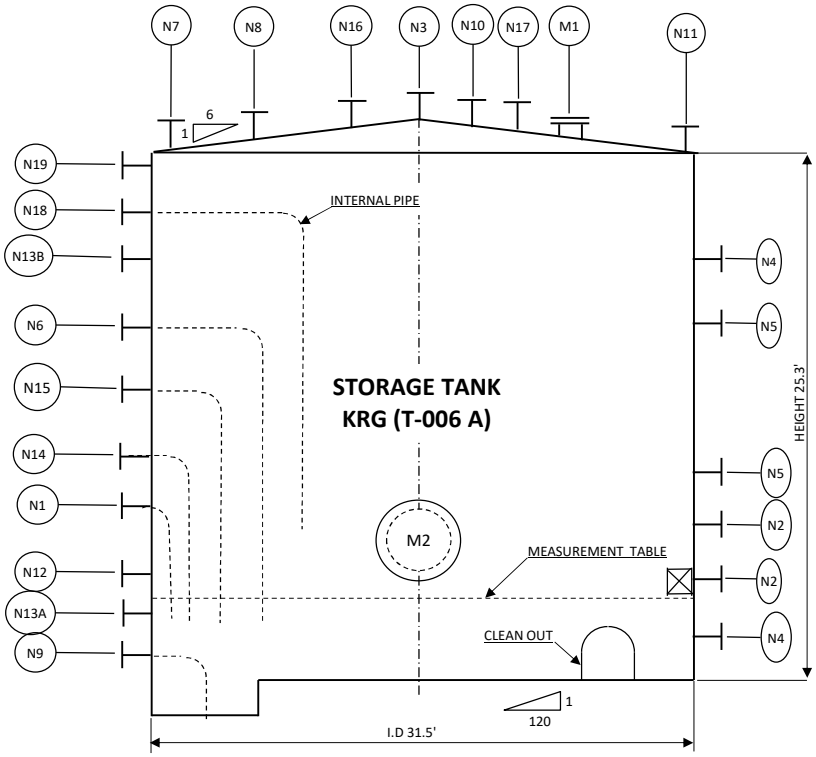
PLATE COURSE (Note 7)

Shell Plate	Height (mm/feet)	C.A (in)	Thickness (mm)	Material	Note
1st	1800/6'	0.125"	8	A283 Gr. C	(Note 22)
2nd	1800/6'	0.125"	8	A283 Gr. C	(Note 22)
3rd	1800/6'	0.125"	6	A283 Gr. C	(Note 22)
4th	1800/6'	0.125"	6	A283 Gr. C	(Note 22)
5th	397/1.3'	0.125"	6	A283 Gr. C	(Note 22)
Bottom Plate	-	0.125"	10	A283 Gr. C	(Note 22)
Roof Plate	-	0.125"	8	A283 Gr. C	(Note 22)

NOZZLE SCHEDULE (Note 8 & 9)

Nozzle Number	Size (in)	Number	Schedule	Flange Rating	Flange Type	Nozzle Projection (mm)	Service	Remarks
SHELL NOZZLE								
N1	6"	1	sch. 80	150#	RF WN	300	Feed Inlet	c/w Internal Pipe (Note 17)
N2	10"	1	sch. 80	150#	RF WN	300	Liquid Otlet	(Note 11)
N4A/B	2"	2	sch. 80	150#	RF WN	250	Level Indicator	
N5A/B	2"	2	sch. 80	150#	RF WN	250	Level Indicator Transmiter	
N6	4"	1	sch. 80	150#	RF WN	300	Minimum Flow	c/w Internal Pipe (Note 17)
N9	4"	1	sch. 80	150#	RF WN	300	Drain Connection	
N12	4"	1	sch. 80	150#	RF WN	300	Future Connection	c/w Blind Flange, Gasket, Bolts & Nuts
N13A/B	2"	2	sch. 80	150#	RF WN	250	Level Indicator	
N14	3"	1	sch. 80	150#	RF WN	300	Flow Inlet from Test Pump KRG	c/w Internal Pipe (Note 17)
N15	3"	1	sch. 80	150#	RF WN	300	Flow Inlet from Water Sump Pit Pump	c/w Internal Pipe (Note 17)
N18	6"	1	sch. 80	150#	RF WN	300	Foam Chamber	
N19	4"	1	sch. 80	150#	RF WN	300	Future Connection	c/w Blind Flange, Gasket, Bolts & Nuts
M2	24"	1	API 650				Manway	c/w Blind Flange, Gasket, Bolts & Nuts
Clean Out	8 x 16	1	API 650				Clean Out	c/w Cover & Davit
ROOF NOZZLE								
N3	4"	1	sch. 80	150#	RF WN	300	Vent (BV)	(Note 25)
N7	2"	1	sch. 80	150#	RF WN	250	Overflow	c/w Blind Flange, Gasket, Bolts & Nuts
N8	16"	1	sch. 80	150#	RF WN	300	ERV	
N10	8"	1	sch. 80	150#	RF WN	300	Gauge Hatch	(Note 12)
N11	2"	1	sch. 80	150#	RF WN	250	Pressure Indicator	
N16	2"	1	sch. 80	150#	RF WN	250	Spare	c/w Blind Flange, Gasket, Bolts & Nuts
N17	2"	1	sch. 80	150#	RF WN	250	Spare	c/w Blind Flange, Gasket, Bolts & Nuts
M1	24"	1	API 650				Manway	c/w Blind Flange, Gasket, Bolts & Nuts

SKETCH



REVISION

NOTES

VTC=Vendor to Confirm , VTA=Vendor to Advise, TBC = To be confirmed , N/A= Not Applicable , C.A= Corrosion Allowance, EQ= Equal

1. Supplier/Vendor shall provide design, supply, fabrication, installation, inspection and testing of Test Tank shall comply with the following:
   - Doc No. TMBP-SP001-ME-SPE-D102, Specification for Tank;
   - Doc No. TMBP-SP001-PR-PID-106, P&ID Storage Tank KRG (T-006 AB)
   - Doc No. TMBP-SP001-PR-DST-D115, Process Datasheet - Storage Tank KRG (T-006 AB)
   - Doc No.TMBP-SP001-PR-CAL-D119, Storage Tank KRG (T-006 AB)
2. All the flanges shall be in accordance with ASME B16.5 type, unless otherwise specified. Flanges larger than the scope of ASME B16.5 shall be in accordance with ASME B16.47, series A.
3. Nozzles gasket material shall be 0.175" thick, spiral wound, 316 SS Windings Flexible graphite filter SS inner & outer ASME B16.20. as per Specification for Piping Material Specification Doc. No. TMBP-SP001-PI-SPE-D107.
4. Bolts, Nuts shall be in SF Heavy hex with Fluorocarbon coated, minimum thickness 35 micron
5. Surface preparation and painting shall be as per spec. TMBP-SP001-PI-SPE-D503, Piping Painting & Lettering Specification.
6. Refer to API 650 chap 4.7. Bolting, Separate for anchor, flanges and structural purpose.
7. Thickness indicated as reference, VENDOR shall calculate and confirm the actual thickness after purchase order.
8. Minimum nozzle projection length refer to API STD 650 13th edition and to be finalized by VENDOR.
9. Nozzle Schedule refer to API 650 13th Edition, table 5.6a and 5.7a and to be finalized by VENDOR.
10. Tank gauge and Liquid level refer to TMBP-SP001-PR-DST-D114, "Process Datasheet - Test Tank TMB (T-003) & Test Tank KRG (T-004)"
11. Liquid outlet nozzle is installed minimum as same as measurement table height. Outlet nozzle (N2) shall be completed with vortex breaker.
12. Gauge Hatch with quick open cover shall be inline with measuring table
13. N7 is served as syphone breaker to erase shyphone effect and make sure that the overflow can stop at the desired level
14. VENDOR shall provide Insulation Gasket Kit for connection stainless steel to carbon steel and shall be provided in separate datasheet.
15. The requirement for internal tank cathodic protection shall be analyzed and calculated by VENDOR
16. Roof manhole completed with Emergency Vent.
17. All internal pipe parts are complete with clips and pipe support
18. Stairway, platfrom & ladder shall be supplied by VENDOR
19. Anchor bolt size and qty shall be designed by VENDOR
20. All weld joint on bottom tank and roof shall be tested use vacuum box method.
21. Where blind flanges are specified, they shall be supplied with bolts/nuts, gaskets and spares for construction as follow :
    - Gasket : 200% spares each blind flange nozzle
    - Bolt and Nuts : 10% bolt and 2 nuts (min 2 set each size and type) for each blinded flange nozzle
22. VENDOR shall be provide all required data to obtain MIGAS certification.
23. VENDOR shall be responsible for the full mechanical design and shall confirm all material chosen, thickness, weight, volume and loading.
24. Shall be checked & verified by vendor
25. Storage Tank KRG T-006A is utilizing asset idle with dimension 9.8m (ID) X 7.7 m

MECHANICAL DATASHEET FOR STORAGE TANK KRG (T-006 B)

Judul Proyek	Nomor Dokumen	Rev.	Date	Page
JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR	TMBP-SP001-ME-DST-D112	A	29-Dec-25	13 of 15

Tag No.	:	T-006 B	Client	:	PT. PERTAMINA EP
Service	:	STORAGE TANK KRG	Plant Location	:	SP TANJUNG MIRING
P & ID No.	:	TMBP-SP001-PR-PID-D106	No. of Unit	:	1 (ONE)
Eq. Layout No.	:	-			

REVISION

1	**DESIGN DATA**			**MATERIAL OF CONSTRUCTION**	
2	Fluid	GROSS FLUID (WATER+OIL)		Shell	A283 Gr. C (Note 25)
3	Code	API STD 650 Latest Ed		Roof & Bottom	A283 Gr. C (Note 25)
4	Capacity (Normal/Working)	2020 / 1100.00	bbls	Reinforcing Pads	A283 Gr. C
5	Operating Pressure	0,15	psig	Nozzle Neck - Pipe	A106 Gr.B
6	Operating Temp.	90	°F	Manhole Neck	A283 Gr. C
7	Design Pressure (max.)	Full of water + 1	psig	Manhole Flange	A283 Gr. C
8	Design Pressure (min.)	Vacuum - 0.06	psig	Manhole Cover	A283 Gr. C
9	Design Temperature (max./min.)	150/20	°F	Nozzle Flanges / Blinds	A105
10	Specific Gravity	0,950		Internal Pipe	A106 Gr.B
11	Flowrate IN/OUT	5055/1200	bpd	Coil Heater	A106 Gr.B
12	Type of Tank	Cylindrical Tank		Nozzle Fittings	A234 Gr.WPB
13	Tank Diameter (ID)	31' - 0" (9448)	ft-in (mm)	Bracket Supports	A283 Gr. C
14	Tank Height	16' - 0" (4878)	ft-in (mm)	Roof Structure	A36 / SS400
15	Corrosion Allowance			Earthing Lugs	SS 304
16	Shell/Roof/Bottom	0.125	in	Name Plates & Bracket	SS 316 & A283 Gr. C
17	Joint Efficiency	0,85		Ladder, Handrail, Platform, Circular Stair	A36 / SS400
18	Hydrotest Pressure	Full of Water		Anchor Bolt Chairs	A283 Gr. C
19	Wind Load:			Anchor (By Others) Bolts (Note 6)	A307 Gr. B Hot Dip Galvanized
20	Wind Design	ASCE 7-16		Nuts (Note 6)	A63 Gr. A Hot Dip Galvanized
21	Wind Speed	64,44	mph	Gaskets	**(Note 3)**
22	Exposure	C		Structural Bolts & Nuts	A325 (Hot Dip Galvanized)
23	Importance Factor	1,00		External Bolts / Nuts	A193 Gr. B7 / A194 Gr. 2H **(Note 4)**
24	Seismic Load:	API-650 - ASCE Mapped (Sa & S1)		Internal Bolts / Nuts	A193 Gr. B7 / A194 Gr. 2H **(Note 4)**
25	Seismic Factor	Sa=0.0422 g; S1=0.0627 g; PGA: 0.017 g		Stiffening Ring / Wind Girder	A283 Gr. C / A36 / SS400
26	Class	As per Soil Investigation		Vortex Breaker	A283 Gr. C
27	Importance Factor	1,5			
28	Heaters	N/A			
29	Heating Fluid	N/A			
30	Design Pressure	N/A	psig	**CONSTRUCTION**	
31	Design Temperature (Inlet/Outlet)	N/A	°F	Roof Type	Supported Cone Roof (1:6)
32	Coil Size & Thickness	N/A		Bottom Type	Slope Bottom (1:120)
33	Coil Surface Area	N/A	ft^2	Walkway / Stairway / Pipe Clip	Required
34	Coil minimum length	N/A	ft	Roof Drain System	Not Required
35	**INSPECTION, TESTING & CERTIFICATE**			Vent System	Required
36	Non Destructive Testing:			Firewater Ring and Clip Support	Required
37	Radiography	Yes, as per Code & Spec.		Foam Chamber	Required
38	Ultrasonic	Yes, as per Code & Spec.		Clips and Support	Required
39	Magnetic Partlicle	Yes, as per Code & Spec.		Earthing Boss	Required (2 ea)
40	Dye Penetrant	Yes, as per Code & Spec.		Internal Parts	
41	Hydrotest	As per Code & Spec.		- Vortex Breaker	Required
42	Post Weld Heat Treatment	As per Code & Spec. (if required)		- Internal Pipe	Required
43	Vacuum Test	As per Code & Spec.		- Foam Deflector	Required
44	Material Certification	Yes, Required Type 3.1		- Measurement Table	Required
45	Leak Test /Soap Bubble Test	Yes at 15 psig		- Draw Off Sump	Required
46	MIGAS Certified	Required		**LIQUID LEVEL**	
47	Calibration by DIMET	Required		HHLL (High High Liquid Level)	12.73' ft
48				HLL (High Liquid Level)	12.23' ft
49				LLL (Low Liquid Level)	4.8' ft
50	**PROTECTIVE COATING DATA**			LLLL (Low Low Liquid Level)	2.3' ft
51	Surface Praration Painting	external	Yes (Note 5)		
52		internal	Yes (Note 25)	**ESTIMATED WEIGHTS (NOTE 23)**	
53	Insulation	N/A		Empty	43366 lb 19670 kg
54	Fireproofing	N/A		Operating	652177 lb 295822 kg
55	Cathodic Protection	Yes (Note 15)		Hydrotest	755626 lb 342746 kg
56	Type / Material / Weight	VTA / VTA / VTA			
57					

Any controlled copies may be superseded. Refer to COMPANY Server for controlled version.

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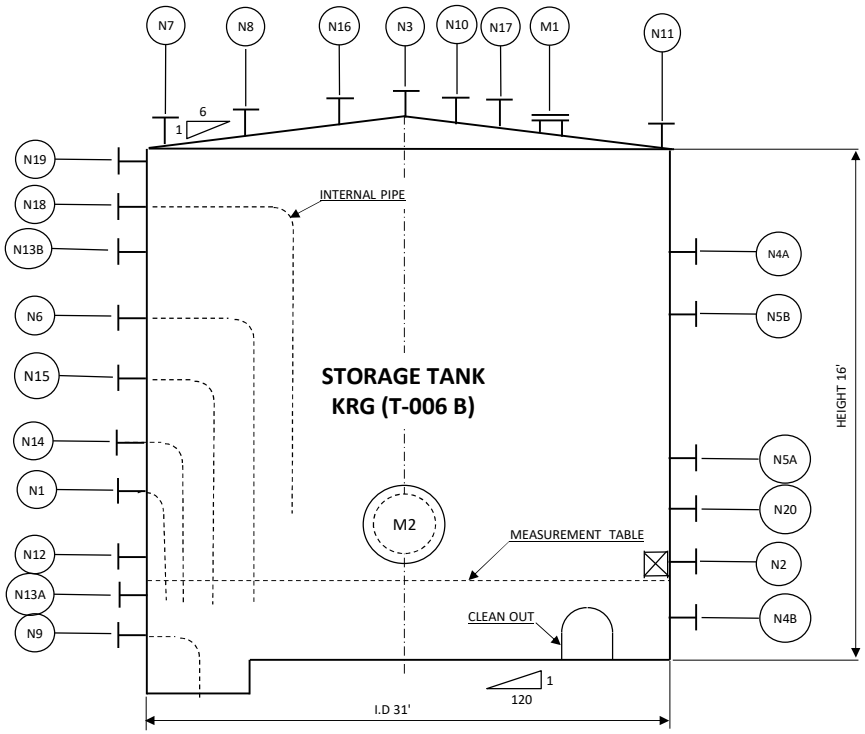
PLATE COURSE (Note 7)

Shell Plate	Height (mm/feet)	C.A (in)	Thickness (mm)	Material	Note
1st	1800/6'	0.125"	8	A283 Gr. C	(Note 22)
2nd	1800/6'	0.125"	6	A283 Gr. C	(Note 22)
3rd	1219/4'	0.125"	6	A283 Gr. C	(Note 22)
Bottom Plate	-	0.125"	10	A283 Gr. C	(Note 22)
Roof Plate	-	0.125"	8	A283 Gr. C	(Note 22)

NOZZLE SCHEDULE (Note 8 & 9)

Nozzle Number	Size (in)	Number	Schedule	Flange Rating	Flange Type	Nozzle Projection (mm)	Service	Remarks
SHELL NOZZLE								
N1	6"	1	sch. 80	150#	RF WN	300	Feed Inlet	c/w Internal Pipe (Note 17)
N2	10"	1	sch. 80	150#	RF WN	300	Liquid Otlet	(Note 11)
N4A/B	2"	2	sch. 80	150#	RF WN	250	Level Indicator	
N5A/B	2"	2	sch. 80	150#	RF WN	250	Level Indicator Transmiter	
N6	4"	1	sch. 80	150#	RF WN	300	Minimum Flow	c/w Internal Pipe (Note 17)
N9	4"	1	sch. 80	150#	RF WN	300	Drain Connection	
N12	4"	1	sch. 80	150#	RF WN	300	Future Connection	c/w Blind Flange, Gasket, Bolts & Nuts
N13A/B	2"	2	sch. 80	150#	RF WN	250	Level Indicator	
N14	3"	1	sch. 80	150#	RF WN	300	Flow Inlet from Test Pump KRG	c/w Internal Pipe (Note 17)
N15	3"	1	sch. 80	150#	RF WN	300	Flow Inlet from Water Sump Pit Pump	c/w Internal Pipe (Note 17)
N18	6"	1	sch. 80	150#	RF WN	300	Foam Chamber	
N19	4"	1	sch. 80	150#	RF WN	300	Future Connection	c/w Blind Flange, Gasket, Bolts & Nuts
M2	24"	1	API 650				Manway	c/w Blind Flange, Gasket, Bolts & Nuts
Clean Out	8 x 16	1	API 650				Clean Out	c/w Cover & Davit
ROOF NOZZLE								
N3	3"	1	sch. 80	150#	RF WN	300	Vent (BV)	(Note 25)
N7	2"	1	sch. 80	150#	RF WN	250	Overflow	c/w Blind Flange, Gasket, Bolts & Nuts
N8	16"	1	sch. 80	150#	RF WN	300	ERV	
N10	8"	1	sch. 80	150#	RF WN	300	Gauge Hatch	(Note 12)
N11	2"	1	sch. 80	150#	RF WN	250	Pressure Indicator	
N16	2"	1	sch. 80	150#	RF WN	250	Spare	c/w Blind Flange, Gasket, Bolts & Nuts
N17	2"	1	sch. 80	150#	RF WN	250	Spare	c/w Blind Flange, Gasket, Bolts & Nuts
M1	24"	1	API 650				Manway	c/w Blind Flange, Gasket, Bolts & Nuts

SKETCH



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NOTES

VTC=Vendor to Confirm , VTA=Vendor to Advise, TBC = To be confirmed , N/A= Not Applicable , C.A= Corrosion Allowance, EQ= Equal

1. Supplier/Vendor shall provide design, supply, fabrication, installation, inspection and testing of Test Tank shall comply with the following:
   - Doc No. TMBP-SP001-ME-SPE-D102, Specification for Tank;
   - Doc No. TMBP-SP001-PR-PID-106, P&ID Storage Tank KRG (T-006 AB)
   - Doc No. TMBP-SP001-PR-DST-D115, Process Datasheet - Storage Tank KRG (T-006 AB)
   - Doc No.TMBP-SP001-PR-CAL-D119, Storage Tank KRG (T-006 AB)
2. All the flanges shall be in accordance with ASME B16.5 type, unless otherwise specified. Flanges larger than the scope of ASME B16.5 shall be in accordance with ASME B16.47, series A.
3. Nozzles gasket material shall be 0.175" thick, spiral wound, 316 SS Windings Flexible graphite filter SS inner & outer ASME B16.20. as per Specification for Piping Material Specification Doc. No. TMBP-SP001-PI-SPE-D107.
4. Bolts, Nuts shall be in SF Heavy hex with Fluorocarbon coated, minimum thickness 35 micron
5. Surface preparation and painting shall be as per spec. TMBP-SP001-PI-SPE-D503, Piping Painting & Lettering Specification.
6. Refer to API 650 chap 4.7. Bolting, Separate for anchor, flanges and structural purpose.
7. Thickness indicated as reference, VENDOR shall calculate and confirm the actual thickness after purchase order.
8. Minimum nozzle projection length refer to API STD 650 13th edition and to be finalized by VENDOR.
9. Nozzle Schedule refer to API 650 13th Edition, table 5.6a and 5.7a and to be finalized by VENDOR.
10. Tank gauge and Liquid level refer to TMBP-SP001-PR-DST-D114, "Process Datasheet - Test Tank TMB (T-003) & Test Tank KRG (T-004)"
11. Liquid outlet nozzle is installed minimum as same as measurement table height. Outlet nozzle (N2) shall be completed with vortex breaker.
12. Gauge Hatch with quick open cover shall be inline with measuring table
13. N7 is served as syphone breaker to erase shyphone effect and make sure that the overflow can stop at the desired level
14. VENDOR shall provide Insulation Gasket Kit for connection stainless steel to carbon steel and shall be provided in separate datasheet.
15. The requirement for internal tank cathodic protection shall be analyzed and calculated by VENDOR
16. Roof manhole completed with Emergency Vent.
17. All internal pipe parts are complete with clips and pipe support
18. Stairway, platfrom & ladder shall be supplied by VENDOR
19. Anchor bolt size and qty shall be designed by VENDOR
20. All weld joint on bottom tank and roof shall be tested use vacuum box method.
21. Where blind flanges are specified, they shall be supplied with bolts/nuts, gaskets and spares for construction as follow :
    - Gasket : 200% spares each blind flange nozzle
    - Bolt and Nuts : 10% bolt and 2 nuts (min 2 set each size and type) for each blinded flange nozzle
22. VENDOR shall be provide all required data to obtain MIGAS certification.
23. VENDOR shall be responsible for the full mechanical design and shall confirm all material chosen, thickness, weight, volume and loading.
24. Shall be checked & verified by vendor